

Q.No.

Name: _____

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Q. 1) Explain the concept of Normalization in DBMS.

Normalization is the process of organizing data in a database to reduce redundancy and eliminate update anomalies. It involves decomposing tables into smaller, well-structured relations.

1NF --- A relation is in 1NF if every attribute contains only atomic values and there are no repeating groups. Each cell must hold a single value.

2NF --- A relation is in 2NF if it is in 1NF and every non-key attribute is fully functionally dependent on the entire primary key. Partial dependencies are removed.

3NF --- A relation is in 3NF if it is in 2NF and no transitive dependency exists. Every non-key attribute must depend directly on the primary key.

BCNF --- A relation is in BCNF if for every functional dependency $X \rightarrow Y$, X must be a super key. BCNF is stricter than 3NF.

Q. 2) What are ACID properties in transaction management?

ACID is an acronym that defines four essential properties of database transactions.

Atomicity --- Either all operations of a transaction execute successfully or none are applied. If a transaction fails midway, all changes are rolled back.

Consistency --- A transaction brings the database from one valid state to another. All integrity constraints and rules must be satisfied before and after the transaction.

Isolation --- Concurrent execution of multiple transactions produces the same result as if they were executed serially. Intermediate states of a transaction are invisible to others.

Durability --- Once a transaction commits, its changes persist permanently in the database even in the event of a system failure.

Q. 3) Write a note on Functional Dependencies.

A functional dependency $X \rightarrow Y$ means that whenever two tuples have the same value for attribute set X , they must also have identical values for attribute set Y .

Armstrong's Axioms

1. **Reflexivity** --- If Y is a subset of X , then $X \rightarrow Y$

2. **Augmentation** --- If $X \rightarrow Y$, then $XZ \rightarrow YZ$

3. **Transitivity** --- If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

These axioms are sound and complete for deriving all functional dependencies implied by a given set.

[illegible]